MATH 7720, HW3, DUE DEC 10

Problem 1, 5pts

Let $\Xi \in \Lambda/(W,\cdot), \lambda^+ \in \Xi$ be the unique element such that $\lambda^+ + \rho$ is dominant and W_0 is the stabilizer of $\lambda^+ + \rho$. Identify center of the category $\mathcal{O}^\Xi$ with $\left(\mathbb{C}[\mathfrak{h}^*]^{coW}\right)^{W_0}$.

Problem 2, 5pts

This problem studies base change for highest weight categories. Let $\mathcal{C}_A$ be a highest weight category, $\tilde{E}$ be the endomorphism algebra of its projective generator $\bigoplus_\tau P_\tau$ as in Lecture 18 (with opposite multiplication so that $\mathcal{C}_A \xrightarrow{\sim} \tilde{E}\text{-mod}_{fg}$). Let A' denote an A-algebra that is a complete local Noetherian ring. Set $\tilde{E}' := A' \otimes_A \tilde{E}$. Define the base change $\mathcal{C}_{A'}$ as $\tilde{E}'\text{-mod}_{fg}$. Show that $\mathcal{C}_{A'}$ is a highest weight category with the same poset as $\mathcal{C}_A$ and the standards $A' \otimes_A \Delta_A(\tau)$.